Aryan Pratap Singh

Software Testing Engineer — Selenium WebDriver, JavaScript, Playwright & pytest

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Summary

Detail-oriented software engineering student with practical experience in test automation using Playwright (browser automation), pytest, and REST API testing in JavaScript, TypeScript, and Python. Experienced in designing test cases, executing regression suites, documenting defects with structured reproduction steps, and collaborating with dev teams on testability. Familiar with C# and Selenium WebDriver concepts. B.Tech IT, PSIT Kanpur (Class of 2027).

Education

Pranveer Singh Institute of Technology — Kanpur, India
Bachelor of Technology in Information Technology — *Oct. 2023 – May 2027*

Jai Narayan Vidya Mandir Inter College — Kanpur, India
*Class XII (CBSE) — **68%*** — *2023*

Jai Narayan Vidya Mandir Inter College — Kanpur, India
*Class X (CBSE) — **88%*** — *2021*

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, C#, Java, C++, SQL

Test Automation: Playwright (Browser Automation / E2E), Selenium WebDriver (Concepts), pytest (Unit & Integration), REST API Testing, WebSocket Validation, Regression Testing, Smoke Testing

Testing Methodologies: Test Case Design, Test Plan Authoring, Defect Lifecycle Management, Root Cause Analysis, Boundary Value Analysis, Equivalence Partitioning, SDLC, Agile/Scrum

Tools: Git, GitHub, Postman, Docker, GitHub Actions (CI/CD), Linux, VS Code

Backend & Databases: Node.js, FastAPI, REST APIs, PostgreSQL, MySQL, MongoDB, SQL Queries

Projects

ThreadLine — Automated E2E Test Coverage & Data Validation thread-line-five.vercel.app | GitHub — *Apr 2026*
Multi-platform data aggregation tool with browser automation and E2E test suites
Next.js 15, TypeScript, Node.js, Playwright, Server-Sent Events, REST APIs

- Wrote Playwright E2E test scripts in TypeScript covering multi-step user flows, async streaming state transitions, input boundary conditions, and error recovery — maintained structured test case documents with intent, steps, expected output, and pass/fail classification.
- Performed regression testing across releases to verify new features did not break existing workflows; tracked defects with steps to reproduce, severity, and environment details.

Mi:RAG — Automated Test Suite for AI Document Pipeline mirag.me | GitHub — *Aug 2026*
Production RAG system with comprehensive pytest automation across all modules
Python, FastAPI, ChromaDB, PyMuPDF, Docker, pytest

- Designed and executed a pytest automation suite covering 6 modules — chunker, embedding, guardrails, vector store, ingestion, and REST API — with unit, integration, regression, and edge-case test scenarios for each.
- Applied equivalence partitioning and boundary value analysis techniques to define test inputs, expected outputs, and failure classifications for document parsing edge cases (empty files, corrupt PDFs, mixed-language content).
- Integrated the test suite into a Docker-based CI pipeline to run automated regression checks on every code change, ensuring release quality standards before deployment.

GitFixAI — REST API Test Automation & Defect Tracking gitfixai.vercel.app | GitHub — *Feb 2026*
Automated code patch tool — Rift Hackathon National Semifinalist (700+ teams)
Python, FastAPI, React, TypeScript, PostgreSQL, WebSockets, GitHub OAuth 2.0

- Wrote integration test cases for all REST API endpoints covering authentication flows, input validation failures, authorization rejection, and WebSocket streaming state verification — documented each defect with environment, steps, and severity.
- Collaborated closely with the development workflow to identify testability gaps during feature design and provide feedback on API contracts and edge-case handling before implementation.

Achievements & Leadership

- **National Semifinalist, Rift Hackathon 2026 (Physics Wallah):** Advanced among 700+ teams across 4 cities with GitFixAI.
- **Top 5%, IntrusionX Hackathon (GLA University, 2026):** Ranked 17th of 354 teams for rapid system development and quality execution.
- **Competitive Programming:** 150+ DSA problems on LeetCode and GeeksforGeeks — arrays, graphs, DP, trees.
- **Event Coordinator, Inter-College Chess Tournament:** End-to-end coordination for 600+ participants.

Certifications

- **DeepLearning.AI:** Retrieval Augmented Generation (RAG) · Agentic AI · Generative AI for Software Development